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arXiv:2502.18625 (q-fin)

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The Market Maker's Dilemma: Navigating the Fill Probability vs. Post-Fill Returns Trade-Off

Jakob Albers, Mihai Cucuringu, Sam Howison, Alexander Y. Shestopaloff

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Using data from a live trading experiment on the Binance Bitcoin perpetual, we examine the effects of (i) basic order book mechanics and (ii) the persistence of price changes from immediate to short timescales, revealing the interplay between returns, queue sizes, and orders' queue positions. We document a fundamental trade-off: a negative correlation between maker fill likelihood and post-fill returns. This dictates that viable maker strategies often require a contrarian approach, counter-trading the prevailing order book imbalance. These dynamics render commonly-cited strategies highly unprofitable, leading us to model `Reversals': situations where a contrarian maker strategy at the touch proves effective.

Subjects: **Trading and Market Microstructure (q-fin.TR)**

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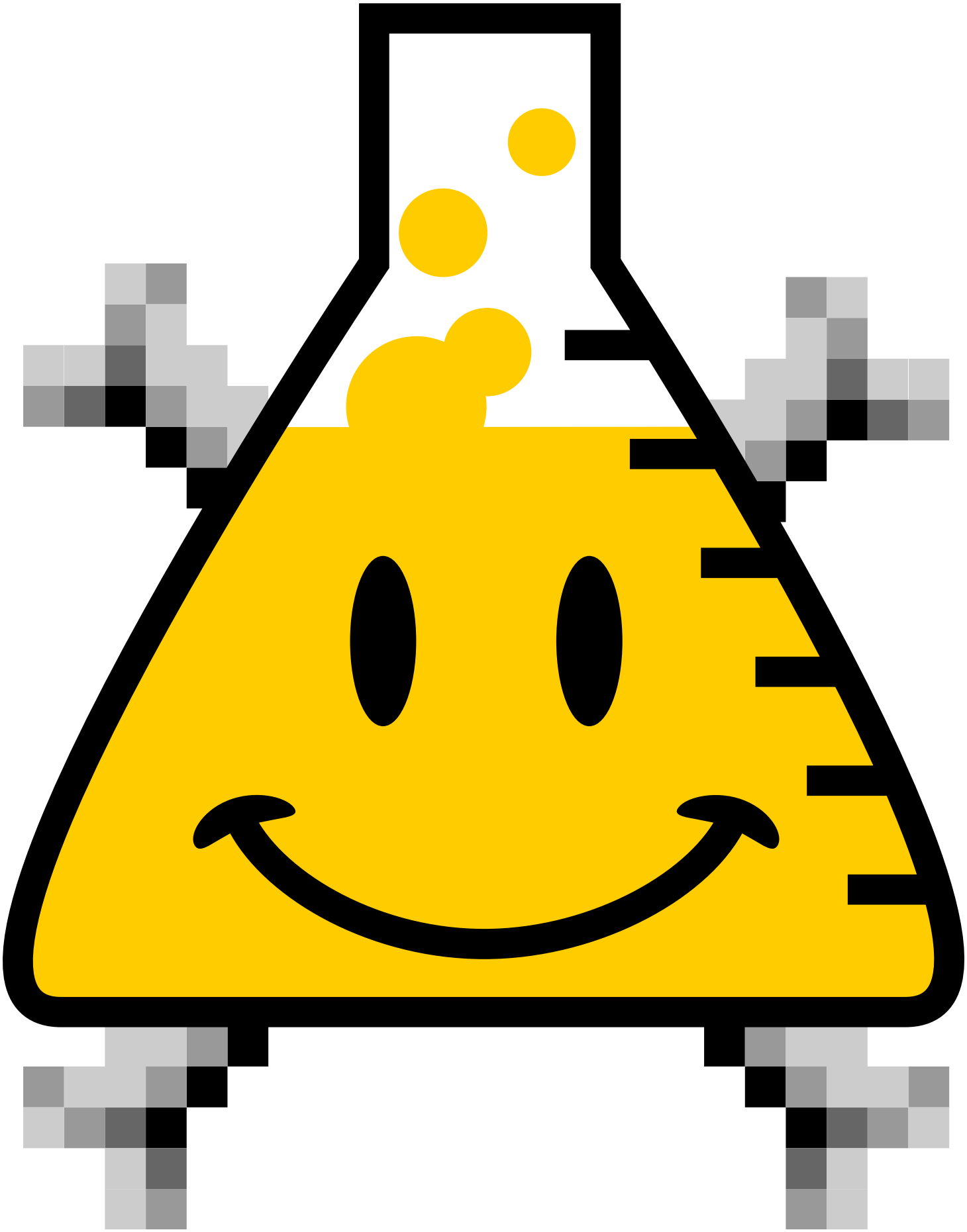

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